Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	*(a)*	(i)		$C_{12}H_{22}O_{11}$ + **18** [O] $\rightarrow$ **6** $(COOH)_2$ + **5** H_2O	1			1		
		(ii)		Mass of the anhydrous acid = 4.05g ∴ Moles of the anhydrous acid = 4.05 / 90 = 0.045 (1) Mass of the water lost = 1.62g ∴ Moles of water = 1.62/18.02 = 0.090 (1) Mole ratio acid : water is 1 :2 Value of x is 2 (1) If no working is shown award (1) for the correct answer		3		3	2	
	(b)			Sample of potassium methanoate not pure / not all HCOOK reacted (1) Inadequate heating / not heated for long enough / not heated at a high enough temperature (1)			2	2		2
	(c)	(i)		Allow to settle / test the filtrate (1) Add a few drops of calcium chloride solution and see if a precipitate forms / cloudiness (1)			2	2		2
		(ii)		Moles of calcium oxalate = 2.49/128 = 0.0195 (1) ∴ Number of moles of potassium oxalate is also 0.0195 Mass of potassium oxalate is 0.0195 × 166 = 3.24 g (1) ∴ % of potassium oxalate in mixture = 3.24 × 100 /4.69 = 69.1 (to 3 sig. figs.) (1) (accept values from 68.9 to 69.1 depending on use of significant figures during the calculation) ecf possible		3		3	1	
				Question 9 total	**1**	**6**	**4**	**11**	**3**	**4**